Notes 3

What is a graphical user interface (GUI)?

A graphical user interface is a set of program that allows a user to interact with the computer system via icons, windows, and various other visual elements.

What is a desktop environment?

A desktop environment provides a predetermined look and feel to the GUI which is broken up to graphical sections and functions such as desktop settings,menus,display manager panels and the list goes on.

What is the command line interface (CLI)?

The user can interact with the computer by inputting commands.

How do I access the command line interface (CLI)?

There are two way to access the CLI which is by the terminal emulator and Linux Console.

What is a virtual console?

A virtual console is a terminal session that runs in Linux System memory. The key combination is CTRL+ALT+ONE OF THE F KEYS

What is a terminal emulator?

A terminal emulator is a program that allows you to access the Linux CLI when using the GUI

What is bash?

Bash is a command line interpreter also known as shell that is used on Unix-like system to interact with the operating system and automate task through scripts.

What is the shell prompt?

A shell prompt will appear whenever the shell is ready to accept input. It typically includes your username followed by current working directory & a dollar sign.

clear

- **Definition**:
  - clears the screen
- **Usage**:
  - `clear` **Example**:
- How to clear the screen:
  - `clear`

echo

Definition:

- Displays text on the screen **Usage:** `echo`+`option`+`string to display` **Example:**
- How to display a line of text:
  - `echo "hello"`
- How to display 2 lines of text
  - `echo -e"hello\world"`

date

Definition:

- Displays or sets the system date and time **Usage:** `date` or `date[format]` **Example:** Typing `date` will show the current date and time.

free

Definition:

- Displays the amount of free and used memory in the system. **Usage:**
  - `free` or `free-h` **Example:**
  - `free-h` will show memory usage in units like KB,MB, or GB.

uname

Definition:

- Displays system information. **Usage:**
  - `uname` or `uname-a` (for all information) **Example:**
  - `uname-a` will output details like the kernel name,network node hostname,kernel release , kernel version, machine hardware name and operating system.

history

Definition:

- Displays a list of commands previously executed in the current shell session. **Usage:**
  - `history` or `history[number]` **Example:**
  - `history 10` will show the last 10 commands executed.

man

Definition:

- Displays the manual page for a given command,providing detailed information about its usage , options, and description. **Usage:**
  - `man[command]` **Example:**
- `man Is` will open the manual page for the `IS` command,explaining its various options and functionalities.

tldr

Definition: Provides simplified,practical examples of command usage. It's a community-driven project aiming to be a "too long;didn't read"version of man pages.

Usage:

- `tldr [command]`

Example:

- `tldr cp` will show common and useful ways to use the cp command with concise explanations.

cheat

Definition:

- Similar to `tldr`,`cheat` provides cheat sheets for command-line tools, offering quick examples and explanations. **Usage:**
  - `cheat[command]` **Example:**
  - `cheat git` will display a cheat sheet for common Git commands.

hostname

Definition:

- Displays or sets the system's hostname

Usage:

- `hostname`

Example:

- Typing `hostname` will print the name of the machine

df

Definition:

- Reports file system disk space usage.

Usage:

- `df` or `df-h`(for human readable output)

Example:

- `df-h` will show the disk space usage for all mounted file systems in a human-readable format

du

Definition:

- Estimates file space usage.It summarizes disk usage of each file argument. **Usage:**
- `du[directory]` or `du -sh[directory]` (fir summary in human-readable format) **Example:**
- `du -sh/home/user/documents` will display the toal size of the `documents` directory in a human-readable format.

figlet

Definition: *creates large letters out of ordinary text,often used for banners or titles. **Usage:**

- `figlet[text]` **Example:**
- `figlet"Welcome"` will display the word "Welcome" in large,stylized ASCII art letters.